

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Final Semester Examination

Summer Semester: 2023 - 2024

Course Code: Chem 4241

Full Marks: 120

Course Title: Chemistry

Time: 2 Hours

There are 4 (four) questions. Answer **all** questions. Marks of each question and corresponding CO and PO are written in the brackets. Programmable calculators are not allowed. Do not write on this question paper. The Symbols have their usual meaning.

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- 1.(a) Explain different concepts of acid-bases with suitable examples. [10] CO2
Discuss the strength of different acids. [5] PO1
- (b) Describe integral methods and half-life method to determine order of the following reaction. [9] CO2
+ PO1
 $A \Rightarrow P$ [6]
- 2.(a) Explain the relationship between temperature and equilibrium constant by deriving van't Hoff's equation. [9] CO1
PO1
A mixture of 41.14 g H₂ and 1325.88 g I₂ vapour is heated in a flask of 1.0 dm³ and 1308.16 HI g is produced at equilibrium. Find out the K_c of the reaction. [6]
[Molecular weight H₂ = 2.016 g, I₂ = 253.8 g and HI 127.908 g]
- (b) Describe collision theory and derive the equation for the rate constant applying the concept of collision theory. [10] CO1
PO1
The rate constant of a first order reaction is $1.05 \times 10^{-2} \text{ s}^{-1}$. If the initial concentration of the reactant is 0.8 mol.dm⁻³. Calculate reaction rate after 3 minutes. [5]
- 3.(a) Explain Faraday's law of electrolysis and mention their applications. [8] CO2
A cell is constructed by following notation. [7] PO1
 $\text{Cd} | \text{Cd}^{2+} (0.001 \text{ M}) || \text{Ag}^+ (0.1 \text{ M}) | \text{Ag}$
The standard reduction potential of Ag⁺|Ag half-cell is 0.80 V and Cd²⁺|Cd is -0.76 V. Calculate cell emf at 25°C. Explain the spontaneity of cell and write down the cell reaction.
- (b) Explain Kohlrausch law of independent migration of ions and write down three applications. [9] CO2
PO1
Derive Nernst equation for determining emf of an electrochemical cell. [6]

- 4.(a) Discuss working principle of a fuel cell and write its advantage. [6] CO1
Calculate (i) degree of dissociation and (ii) hydrogen ion concentration of 0.01 M [4] PO1
propionic acid at 25°C. [$K_d = 1.32 \times 10^{-5}$]
- (b) Draw the phase diagram of water and discuss it with the help of phase rule. [10] CO1
PO1
- (c) Classify the liquid crystals and briefly describe the different types of [10] CO1
thermotropic liquid crystals. PO1

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SEMESTER FINAL EXAMINATION
 DURATION: 2 HOURS

SUMMER SEMESTER, 2023–2024
 FULL MARKS: 120

Phy 4241: Physics II

Please answer according to the order of the questions.

Answer all 4 (FOUR) questions. The symbols have their usual meanings. Include appropriate diagrams and step-by-step calculation where relevant. Marks of each question and the corresponding CO and PO are written in the brackets.

1. a) State and explain Thevenin's theorem. Draw a circuit diagram of your choice and evaluate the Thevenin resistance R_{TH} and the Thevenin voltage E_{TH} at your chosen point in the circuit. 3 + 7
(CO1)
(PO1)
- b) State and explain maximum power transfer theorem. What is the condition for maximum power transfer? From the circuit in Figure 1, with step-by-step calculation, show that the maximum power transfer is given by — 3 + 7
(CO1)
(PO1)

$$P_{I_{max}} = \frac{E_{Th}^2}{4R_{Th}}$$

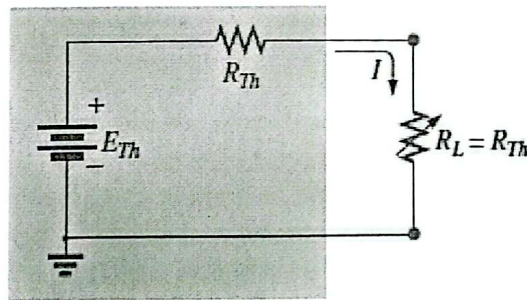


Figure 1: Circuit diagram for Question 1.b

- c) Given the network in Figure 2, find the value of R_L for the maximum power transferred to the load R_L with an appropriately redrawn circuit and step-by-step calculation. 10
(CO1)
(PO1)

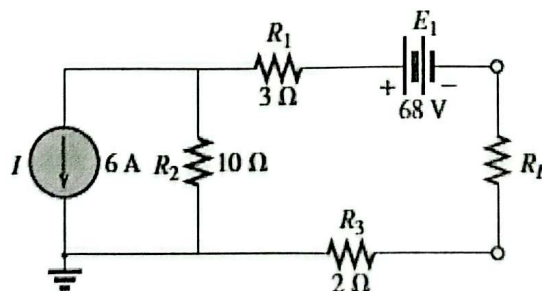


Figure 2: Network diagram for Question 1.c

2. a) What is source transformation? With the help of a circuit diagram, describe an independent and a dependent source. 3 + 7
(CO1)
(PO1)

- b) Use source transformation to find v_o in the circuit of Figure 3 with an appropriately redrawn diagram.

10
(CO2)
(PO1)

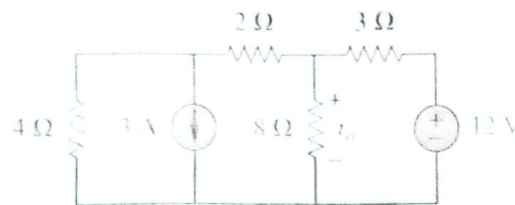


Figure 3: Circuit diagram for Question 2 b

- c) Describe Delta-Y (wye) and Y-Delta transformation and its importance in solving electrical circuit problem. Draw a suitable circuit diagram and derive the equations for converting an Y to Delta and vice-versa. 4 + 6
(CO2)
(PO1)
3. a) Describe a sinusoidal waveform with a diagram highlighting all its parameters. Draw the circuit diagrams with corresponding sinusoids for an ac circuit with (i) a resistive, (ii) an inductive, and (iii) a capacitive load. Obtain an equation for the total impedance Z for all three passive elements connected in series with the ac source with appropriate circuit and sinusoid diagrams. 2 + 3 + 5
(CO2)
(PO1)
- b) With an appropriate diagram, describe the operation of an LCR circuit. Define Impedance Z highlighting the meaning of capacitive reactance and the inductive reactance. 5 + 5
(CO2)
(PO1)
- c) Draw the sinusoids of an inductive and capacitive ac circuit showing the corresponding phases of current and voltages. Describe how the electrical and magnetic energies are built up in the capacitor and the inductor depending on the phase differences between the current and voltage with appropriate diagrams. 4 + 6
(CO2)
(PO1)
4. a) Describe the origin of magnetism in a material. Distinguish magnetic reluctance and electrical resistance. Draw a neat magnetic circuit diagram and its corresponding electric analog. Elaborate on the two different driving forces in these two circuits in the light of Faraday's law of electromagnetic induction. 3 + 7
(CO2)
(PO1)
- b) Find the value of I required to establish a magnetic flux $\phi_B = 0.75 \times 10^{-4} \text{ Wb}$ in the series magnetic relay circuit. (Given Area = $1.5 \times 10^{-4} \text{ m}^2$, $l_{cdefab} = 100 \times 10^{-3} \text{ m}$ and $l_{bc} = 2 \times 10^{-3} \text{ m}$, $N = 200$ turns) 5 + 5
(CO2)
(PO1)

Draw the Magnetic circuit and its equivalent electric circuit analogy for the relay in Figure 4.

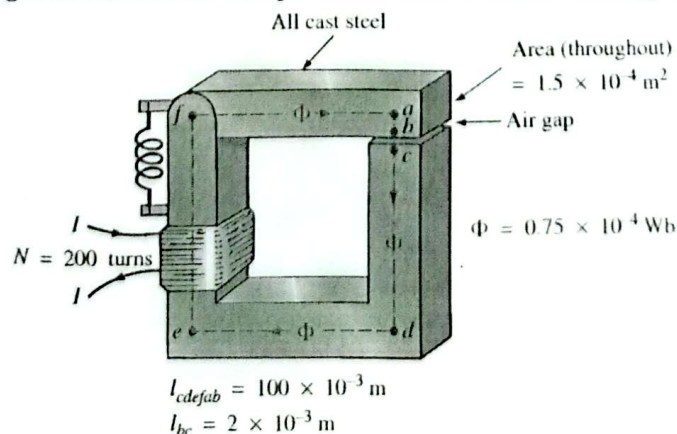


Figure 4: Relay image for Question 4.b

- c) Describe Phasor and draw the phasor representation of two sinusoidal waveforms. Describe how you would convert a sinusoid from the time domain to the phasor domain. Can you apply phasor algebra to the waveforms of two different frequencies? Justify. 5 + 5
(CO2)
(PO1)

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SEMESTER FINAL EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 120

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper. Show all necessary intermediate steps to support the correctness of your answer. Use a pencil for drawing diagrams, as it allows for easier corrections.

Answer all 4 (four) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
1. a) A combinational circuit is specified by the following functions: 10
(CO2)
(PO3)

$$F_1(a, b, c) = (3, 5, 6, 7)$$

$$F_2(a, b, c) = (0, 2, 4, 7)$$

Design a $3 \times 4 \times 2$ Programmable Logic Array (PLA), showing the connection of input variables, product terms, and output functions following appropriate intermediate steps
 - b) Determine the required ROM size to implement the following circuits with proper justifications: 2.5 × 4
 - i. A BCD adder/subtractor with a control input to select between addition and subtraction. (CO1)
(PO1)
 - ii. A binary multiplier that multiplies two 4-bit numbers.
 - iii. Dual 4-to-1 line multiplexers with common selection inputs.
 - iv. A 4-bit magnitude comparator.
 - c) Implement a T flip-flop using an SR flip-flop, showing all intermediate steps. 10
(CO1)
(PO1)
-
2. a) Design 4-bit synchronous Ring and Johnson counters using JK flip flops with positive-edge triggered clocks including the following tasks: 4+4+
6+6
(CO1)
(PO1)
 - i. Draw the complete timing diagrams for all valid states of each counter.
 - ii. List all used and unused states for each counter and briefly explain the method for recovering the unused states.
 - iii. For each counter, derive the minimized state expressions for all valid states, using the minimum number of variables.
 - b) Design a MOD-10 synchronous binary up-down counter using T flip-flops. 10
(CO4)
(PO1)
-
3. a) A seven-segment display is an electronic device consisting of seven individually controlled Light-Emitting Diodes, labeled 'a', 'b', ..., 'g', arranged to display decimal digits (0 – 9) shown in Figure 1. A BCD-to-seven segment decoder accepts a 4-bit Binary Coded Decimal (BCD) input and generates the appropriate seven control signals to illuminate the required segments. 12 + 3
(CO3)
(PO1)

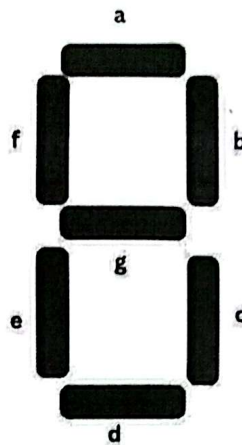


Figure 1: A seven-segment display for Question 3.a

- i. Design a BCD-to-seven segment decoder using 4-to-1 multiplexers, clearly showing all intermediate steps.
- ii. Apply dont-care conditions (×) effectively to minimize the complexity of the implementation.

b) Design a sequential circuit described by the following state equations using JK flip-flops:

$$A_{(t+1)} = xAB + yA'C + xy$$

$$B_{(t+1)} = xAC + y'BC'$$

$$C_{(t+1)} = x'B + yAB'$$

15
(CO3)
(PO3)

4. a) Design a sequence generator to produce the bit sequence 111011 using JK flip-flop. Show all intermediate steps in detail including:
 - i. Identify the minimum number of flip-flops required to represent the states.
 - ii. Construct the state diagram for the derived sequential circuit, clearly showing all possible states and transitions.
 - iii. Identify and justify the type of this sequential circuit with appropriate reasoning.
- b) Design a 3-bit Gray code down counter using JK flip-flops, showing all intermediate steps.

8+2+
3+2
(CO2)
(PO1)

15
(CO2)
(PO1)

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 DURATION: 2 HOURS

SUMMER SEMESTER, 2023-2024
 FULL MARKS: 120

CSE 4203: Discrete Mathematics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 4 (four) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

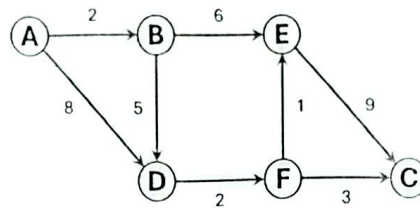


Figure 1: Directed Weighted Graph for Question 1.a and Question 1.b

1. a) Illustrate how Dijkstra's algorithm can be applied to find the shortest path from node A to node C of the directed weighted graph in Figure 1. 10
(CO2)
(PO1)
 - b) Given the directed weighted graph in Figure 1, starting from vertex A, perform both *Breadth-First Search* (BFS) and *Depth-First Search* (DFS) to generate the respective directed spanning trees with a sequential illustration of the process from start to end. Visit the unvisited vertices in alphabetical order when multiple choices are available. 5 + 5
(CO1)
(PO1)
 - c) "Dijkstra's Algorithm will work only for directed and undirected graphs with positive weights," explain with a suitable example. 5
(CO1)
(PO1)
 - d) Apply Dirac's and Ore's theorem to the graph in Figure 1 and determine the presence of a Hamiltonian circuit while ignoring the directions. 5
(CO3)
(PO2)
 - e) Explain the concept of *Relaxation* in Dijkstra's Algorithm with a suitable example. 5
(CO1)
(PO1)
2. a) Perform a two-step encryption on the message, "THIS MARKS THE END OF DISCRETE MATHEMATICS"; firstly, by using public key cryptography with a key value of $k = 5$, and then by using transposition cipher with $\sigma = \{1, 2, 3, 4\}$ and $\sigma(1) = 2, \sigma(2) = 4, \sigma(3) = 1$, and $\sigma(4) = 3$. Ignore spaces during the encryption process. 5 + 5
(CO2)
(PO1)
 - b) Use Huffman Coding to generate the prefix codes of the characters in the string, "THIS MARKS THE END OF DISCRETE MATHEMATICS" Calculate the average number of bits required to represent each character using the generated prefix codes. 15 + 5
(CO2)
(PO1)

[N.B.: Include spaces and consider arranging characters/sub-trees with the same probability in lexicographic order. For equal probabilities, assign the character/sub-tree that appears earlier as the left child in the next iteration.]
 - c) How does Huffman coding resolve the issues associated with simple prefix codes? Explain with an example. 5
(CO2)
(PO1)

3. A manager at a pizza joint must decide how long to promise customers their orders will take as the number of deliveries, x , grows. The pizza shops delivery time for an order of x pizzas is modeled by $f(x) = 7x^2$ (minutes). Based on the delivery time of the joint in recorded history, Connie, a very conservative manager, says, "Your pizza might take up to $g(x) = x^3$ minutes." On the other hand, Leo, an optimistic manager of the joint, says, "Your pizza will take about $h(x) = 10x^2$ minutes." Based on this information, answer the following —

a) Examine if Connie's and Leo's estimates provide valid upper bounds on the actual delivery time. Include a mathematical justification for each estimate.

7 + 7
(CO2)
(PO1)

b) Determine whether Leo's estimate $h(x)$ is a tight bound for the actual delivery time $f(x)$.

6
(CO2)
(PO1)

c) As the number of deliveries becomes very large, which estimate becomes less practical? Explain using the concept of growth rates.

5
(CO2)
(PO1)

4.

$$\frac{\sqrt{b^2 - 4ac}}{2ac - b} \quad (1)$$

Answer the following questions based on Equation 1 —

a) Generate the corresponding rooted tree.

5
(CO1)
(PO1)

b) Generate the corresponding *prefix* and *postfix* notations.

5 + 5
(CO2)
(PO1)

c) Evaluate the *prefix* and *postfix* notations considering $a = 2, b = 5, c = 2$.

5 + 5
(CO2)
(PO1)

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DEPARTMENT OF NATURAL SCIENCE

Semester Final Examination

Summer Semester: 2023-2024

Course Code: **MATH 4241**

Full Marks: 160

Course Title: **Integral Calculus and Differential Equations**

Time: 2 Hours

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 4 (four) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. (a) Define a Partial Differential Equation (PDE). Form a PDE by eliminating the arbitrary constants from $z = Ae^{-\alpha x} \cos \beta x \sin \gamma y$ with $\alpha^2 = \beta^2 + \gamma^2$. [10]
(CO1)
(PO1)
- (b) Find the general solution of the PDE $y^2(x-y)p + x^2(y-x)q = z(x^2 + y^2)$ by Lagrange's method. [15]
(CO2)
(PO2)
- (c) Find the integral surface of the PDE $2y(z-3)p + (2x-z)q = y(2x-3)$ through the curves $x^2 + y^2 = 2x$, $z = 0$. [15]
(CO3)
(PO3)
2. (a) When a system of first order equations is said to be compatible? Discuss briefly. [6]
(CO1)
(PO1)
- (b) Show that the two functions represented by $f(x, y, z, p, q) = xp - yq - x = 0$ and $g(x, y, z, p, q) = x^2p + q - zx = 0$ are compatible. Hence, find their solution. [16]
(CO4)
(PO1)
- (c) Define complete integral. Find a complete integral of $(p^2 + q^2)y = qz$ using Charpit's method. Hence, obtain its singular integral (if exists) and general integral. [18]
(CO4)
(PO1)
3. (a) Solve $y'' - 4y' + 4y = (x+1)e^{2x}$. [10]
(CO1)
(PO1)

- (b) Use the substitution $x = e^t$ to transform $x^2 y'' - xy' + y = \ln x$ to a differentiable equation with constant coefficients and then solve it. [15]
(CO2)
(PO1)
- (c) After a mass weighing 10 pounds is attached to a 5-foot spring, the spring measures 7 feet. This mass is removed and replaced with another mass that weighs 8 pounds. The entire system is placed in a medium that offers a damping force numerically equal to the instantaneous velocity. Find the equation of motion if the mass is initially measured from a point $\frac{1}{2}$ foot below the equilibrium position with a downward velocity of 1 ft/s. [15]
(CO3)
(PO3)
4. (a) Use the method of Frobenius to obtain two linearly independent series solutions about $x = 0$ of $2xy'' - y' + 2y = 0$. Form the general solution on the interval $(0, \infty)$. [15]
(CO1)
(PO1)
- (b) Use the recurrence relation to express $J_{3/2}(x)$ in terms of $\sin x$, $\cos x$ and powers of x . [10]
(CO2)
(PO2)
- (c) The differential equation $(1-x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1-x^2}\right]y = 0$ is known as the associated Legendre equation. When $m=0$ this equation reduces to Legendre's equation. A solution of the associated equations is $P_n^m(x) = (1-x^2)^{\frac{m}{2}} \frac{d^m}{dx^m} P_n(x)$, where $P_n(x)$, $n=0,1,2,\dots$ are the Legendre polynomials and the solutions $P_n^m(x)$ for $m=0,1,2,\dots$ are called associated Legendre functions. [15]
(CO4)
(PO1)
- (i) Find the associated Legendre functions $P_0^0(x)$, $P_1^0(x)$, $P_1^1(x)$, $P_2^1(x)$.
- (ii) What can you say about $P_n^m(x)$ when m is an even non-negative integer?

The End

Program: B.Sc.C.S.E 2nd Semester
Semester: Summer

Date: September 12, 2025 (Friday)
Time: 10:00 AM – 12:00 PM

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF TECHNICAL AND VOCATIONAL EDUCATION (TVE)

Examination: Semester Final Examination
Course Number: Hum 4241
Course Title: Islamic History Science and Culture

Academic Year: 2023 - 2024
Full Marks: 80
Time: 2.00 Hours

There are 4 (four) questions. Answer all of them. The symbols have their usual meaning.

1. a) If Al-Khwarizmi's "al-jabr" shaped algebra, how does his legacy live on every time you use an algorithm in coding? 10 CO2 PO2
b) What role did the introduction of zero and Indian numerals play in shaping modern computer engineering? 10 CO1 PO1
2. a) Al-Biruni estimated Earth's radius within 50 km accuracy. Explain how his triangulation methods could guide modern satellite engineering. 10 CO4 PO8
b) The Qur'an describes the division of the atom. Discuss the lessons that nuclear engineers can learn from this early reference. 10 CO3 PO6
3. a) If Al-Mamun's "House of Wisdom" was a hub of global knowledge exchange, what would its equivalent look like in today's digital age of engineering? 10 CO2 PO2
b) In the light of lā ḍarar wa lā ḍirār ("no harm, no reciprocating harm"), explain how you would decide on a project that risks environmental damage. 10 CO3 PO6
4. a) Discuss the role that young Muslim engineers should play in turning 'brain drain' into 'brain gain' for their nations. 10 CO1 PO1
b) If the Muslim Golden Age once lit the world with innovation, how can today's Engineers rekindle that flame in AI and renewable energy? 10 CO4 PO8